

Syllabus

UG0802 -ZOO-52T-103- Diversity of Chordates & Developmental Biology of Vertebrates

UG0802 -ZOO-52P-104- Practicals based on Diversity of Chordates & Developmental Biology of Vertebrates

II-Semester -Zoology

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
II	ZOO-52T-103 ZOO-52P-104	Diversity of Chordates & Developmental Biology of Vertebrates Practicals based on Diversity of Chordates & Developmental Biology of Vertebrates			5	6
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
5	Major	4	2	6	Yes	Lectures: 60 lectures including diagnostic and informative assessments during lecture hours and 30 Hours of Practical training/demonstration.
List of Program Codes in which Offered as Minor Discipline		B.Sc. Chemistry: UG0804 B.Sc. Botany: UG0805				
Prerequisites		B.Sc. I Semester (Bio Group)				
Objectives of the Course:		- The course offers a complete understanding about diversity and classification of Chordate animals. - It educates the students regarding general and specific characteristics of chordates. Thorough understanding of their affinities and evolutionary aspects of chordates will be developed in students. - The course will also provide a glimpse of scope and historical background of developmental biology to the students. - It will impart knowledge regarding basic concepts of differentiation, morphogenesis and pattern formation and insight into stem cells and cloning. - Understanding of essential events of developmental biology will be imparted through proper explanation of gametogenesis, stages of embryonic development and foetus formation.				

Detailed Syllabus

ZOO-52T-103: Diversity of Chordates & Developmental Biology of Vertebrates

Unit-I

General characteristics and classification of Chordata up to classes; Urochordata: General characteristics and classification up to sub-classes, Study of *Herdmania* and its affinities. **4 Hrs**

Cephalochordata: General characteristics and classification up to sub-classes, Study of *Branchiostoma*(Amphioxus) and its affinities **4 Hrs**

Cyclostomata (Agnatha): General characteristics and classification up to sub-class; Study of *Petromyzon* and its affinities **3Hrs**

Pisces: General characteristics and classification up to sub-classes; Types of fins and scales, swim bladder, Weberian ossicles, Parental care and migration in fishes. **4Hrs**

Unit-II

Amphibia: General characteristics and classification up to sub-classes; origin and evolution of Amphibia; Neoteny; Parental care in Amphibians. **4 Hrs**

Reptilia: General characteristics and classification up to sub-classes; Identification of poisonous and non-poisonous snakes. **3Hrs**

Aves: General characteristics and classification up to sub-classes; Flight adaptations and Migration in birds. **4 Hrs**

Mammals: General characteristics and classification up to sub-classes; Dentition in Mammals; Adaptive radiation in mammals. **4 Hrs**

Unit-III

Scope and History of Developmental Biology; Early Embryonic Development: Gametogenesis: Spermatogenesis and Oogenesis; **4 Hrs**

Fertilization; Cleavage: planes and patterns of cleavage; blastulation and morulation; parthenogenesis **3Hrs**

Gastrulation: Types of morphogenetic movements; Embryonic induction; Fate of germ layers, Fate maps **4 Hrs**

Early embryonic development of frog (up to neurulation) and chick (up to 96 hrs). **4Hrs**

Unit-IV

Metamorphosis and its hormonal regulation in frogs; Regeneration of limb in frog **5Hrs**

Types and functions of extra embryonic membranes in chick development **2Hrs**

Types, formation and functions of placenta in mammals, Implantation, Pregnancy and Parturition **5 Hrs**

Teratology and Developmental disorders. **3 Hrs**

Suggested Books and References:

1. Biology. Campbell & Reece (2005)., Pearson Education, (Singapore) Pvt. Ltd.
2. Chordate Zoology. Jordan E.L., Verma P. S. (2022) S. Chand and Company Limited.
3. Biology, 6th edition. Raven, P. H. and Johnson, G. B. (2004) Tata McGraw Hill Publications. New Delhi.
4. Analysis of Vertebrate Structure. Hilderbrand, M and Gaslow G.E.. John Wiley and Sons
5. Principles of Developmental Biology (4th edition). Wolpert, L & Tickle, C (2011). Oxford University Press, ISBN: 9780198792918
6. Patten's Foundations of Embryology. Carlson, Bruce M (1996). McGraw Hill, Inc. ISBN: 9780070634275
7. The Life of Vertebrates. III Edition. Young, J. Z. (2004) Oxford university press.
8. Comparative Anatomy and Development Biology of Vertebrates (2024) Dr Jyotsna Jain, Dr Dev D. Patel, Dr Pallavi Kaushik and Dr Dau Lal Bohra. Text book for B.Sc. II Semester, Neelkanth Publishers (P) Ltd, Jaipur, India 2024 ISBN: 978-93-5736-733-2.
9. Developmental Biology. X Edition. Gilbert, SF (2014) Sinauer Associates, Inc., Publishers, Sunderland, Massachusetts, USA. ISBN : 9780878939787
10. An Introduction to Embryology. Balinsky, B.I. (2008). International Thomson Computer Press.

Suggested E-resources:

1. Meena G, 2020. Developmental Biology, Glossary, Ideal International Publication Pvt.Ltd.
<https://drive.google.com/file/d/1ebK1B6QHc6fJG6CXaGicmXTZlY6VkOxi/view?usp=drivesdk>
2. <https://innovativezoology.uniraj.edu.in/>

Course Learning Outcome:

Upon completion of this course, students will be able to:

- Know about the levels of organization among different groups of vertebrates.
- Understand how chordates evolved during the course of evolution through succession.
- Know the evolution of different concepts in developmental biology.
- Understand the process of gamete formation from stem cell population to mature ova and sperm.
- Comprehend the sequence of steps leading to the formation of gametes and development of embryo.
- Know the mechanisms underpinning cellular diversity and specificity in animals.
- Have the knowledge about the methods and tools related to developmental biology which help to understand different processes of embryogenesis.

Practical Syllabus

UG0802 -ZOO-52P-104- Practicals based on Diversity of Chordates & Developmental Biology of Vertebrates

1. Anatomy: Study of swim bladder and Weberian ossicles in any edible fish.
2. Study of microscopic slides and museum specimens:
 - i. **Protochordates:** *Herdmania*, *Ciona*, *Botryllus*, *Amphioxus*, *Doliolum*, *Oikopleura*, *Pyrosoma*, Tadpole larva of *Ascidia*
 - ii. **Agnatha:** *Petromyzon*, *Myxine*, *Ammocoete* larva.
 - iii. **Pisces:** *Zygaena* (*Sphyrna*), *Torpedo*, *Pristis*, *Chimaera*; *Acipenser*, *Amia* or *Lepidosteus*, *Labeo*, *Clarias*, *Anguilla*, *Hippocampus*, *Exocoetus*, *Echenies*, any flat-fish, *Syngnathus*, *Protopterus*, *Lepidosiren*, *Neoceratodus*, *Notopterus*.
 - iv. **Amphibia:** *Ichthyophis*, *Necturus*, *Proteus*, *Ambystoma*, *Salamander*, *Axolotl*, *Siren*, *Alytes*, *Hyla*, *Pipa*, *Rachophorus*, *Rana*
 - v. **Reptilia:** *Testudo*, *Chelone* and fresh water tortoise, *Sphenodon*, *Hemidactylus*, *Phrynosoma*, *Draco*, *Calotes*, *Chameleon*; *Eryx*, *Hydrophis*, *Krait*, *Naja*, *Viper*, *Bungarus*, *Crocodilus*, *Alligator*.
 - vi. **Aves:** *Pavocristatus* (peacock), *Choriotis* (Great Indian Bustard), *Columba*, *Archaeopteryx*
 - vii. **Mammalia:** *Ornithorhynchus*, *Echidna*, *Tachyglossus*, *Didelphys*, Kangaroo, Bat, *Loris*, *Manis*, Mongoose, Otter
3. Study of the following through Permanent Slide preparations: oral hood of amphioxus, scales of fishes, hair of mammals
4. Frog - Study of developmental stages - through permanent slides (whole mounts and sections) — cleavage stages, blastula, gastrula, neurula, tail bud stage, tadpole external and internal gill stages.
5. Study of Chick Embryo: 18 hrs, 21 hrs, 24 hrs, 33 hrs, 48 hrs, 72 hrs and 96 hrs of incubation.
6. Window making in chick eggs to study the various incubation stages of developing chick embryo
7. Study of extra-embryonic membranes in chick development.
8. Educational tour: Visit to Zoo/National Park/Sanctuary and submission of report.

Scheme of Practical Examination and Distribution of Marks

S.No.	Practical Exercise	Regular Students	Ex. /N.C. Students
1.	Major exercise	8	12
2.	Permanent slide preparation	7	6
3.	Developmental Biology	7	6
3.	Identification and comments on Spots (1 to 8) Note: Spot 1 to 4 from Chordata and 5 to 8 from Developmental Biology.	16	16
4.	Viva Voce	7	10
5.	Class Record	5	
		50	50

Note:

*Internal marks for regular students only.

1. Anatomy: Study of systems of the prescribed types with the help of dissection. Detailed charts/Dissection softwares/virtual tools/models can also be utilized to study anatomy.
2. With reference to microscopic slides, in case of non-availability, the exercise should be substituted with diagrams / photographs.
3. Candidates must keep a record of all work done in the practical class and submit the same for inspection at the time of the practical examination.
4. Mounting material for permanent preparations would be as per the syllabus or as available through collection and culture methods.
5. It should be ensured that animals used in the practical exercises are not covered under the wild life act 1972 and amendments made subsequently.

Syllabus

UG0802 -ZOO-63T-201- Economic Zoology & Ethology

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